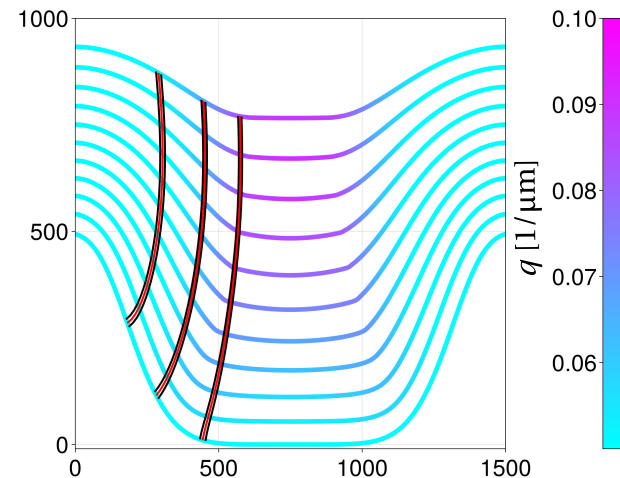
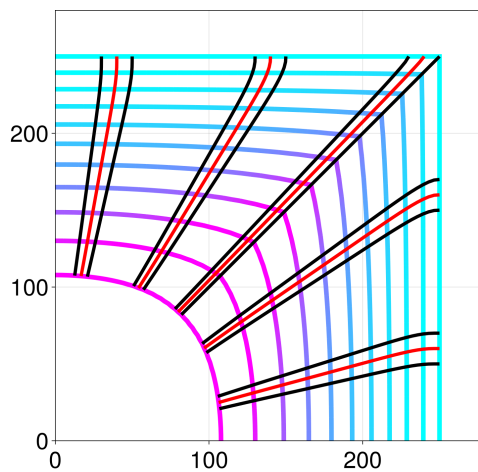
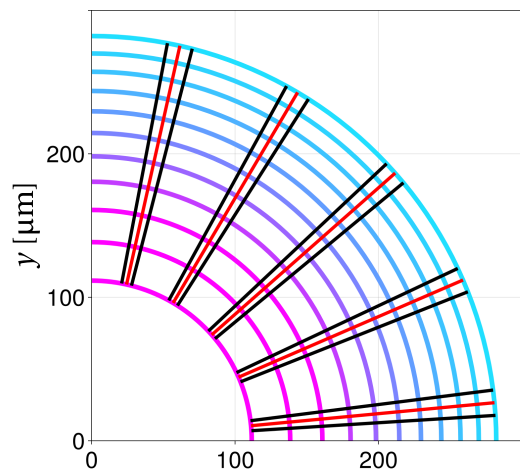


a)

Linear force: $f(\ell) = k_0 (\ell - a_0)$ 

b)

Nonlinear force: $f(\ell) = k_1 \left(\frac{1}{a_1} - \frac{1}{\ell} \right)$ 